

PROBLEM

PROBLEM 1: WATER JUG PROBLEM

Problem statement:

You are given two water jugs with capacities of 4 gallons and 3 gallons. Neither jug has measurement markings. Using only the operations of filling a jug, emptying a jug, and pouring water from one jug to the other, determine a sequence of actions to obtain exactly 2 gallons of water in the 4-gallon jug.

PROBLEM 2: MARS ROVER

Problem statement:

A Mars Rover is an intelligent autonomous agent that explores the Martian surface, collects and analyses rock and soil samples, and sends scientific data back to Earth. Analyse the Mars Rover as an intelligent agent by identifying its percept's operating environment, actions, performance measures, and the most suitable agent architecture.

PROBLEM 3: 8-QUEENS PROBLEM

Problem statement:

Place 8 queens on an 8×8 chessboard such that no two queens attack each other. This means that no two queens can share the same row, column, or diagonal. Formulate the problem by defining the search space, initial state, goal state, operators, and path cost using AI search strategies.

PROBLEM 4: OLA CAB BOOKING

Problem Statement:

A customer wants to travel from one location to another using the OLA Cab booking application. The customer can choose different cab types such as Mini, Micro, Sedan, Shared, and Prime based on comfort and travel requirements. Identify the appropriate AI problem-solving agent and write the pseudocode for the cab selection process.

PROBLEM 5: UNIFORM COST SEARCH (UCS)

Problem Statement:

A logistics company operates a delivery network in which packages must be transported between warehouses at the minimum possible cost. Each route has an associated travel cost. Using the given weighted graph, apply the Uniform Cost Search (UCS) algorithm to determine the least-cost path from the starting warehouse S to the destination warehouse G.